

MINISTRY OF EDUCATION AND TRAINING
EASTERN INTERNATIONAL UNIVERSITY



MIS 395
ARTIFICIAL INTELLIGENCE FOR BUSINESS

Final Project

**Multidomain Analytics for Strategic
Decision-Making**

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I. Project Motivation

As artificial intelligence (AI) continues to transform the business landscape, its integration into strategic decision-making processes is becoming increasingly essential. This project is motivated by the need to harness AI-powered tools across diverse business domains - namely marketing, retail operations, and human resources - to uncover actionable insights and drive performance improvements.

In the marketing domain, we utilize the Bank Marketing dataset from the UCI Machine Learning Repository, which provides detailed records of direct marketing campaigns. The application of AI to this data enables organizations to predict customer responses, enhance targeting strategies, and improve campaign effectiveness.

In the realm of retail analytics, the Bike Sales dataset from Kaggle offers over 100,000 transaction records. AI techniques applied to this data allow businesses to detect sales patterns, forecast demand, and optimize inventory planning - critical factors in maintaining operational efficiency and profitability.

In human resource management, the Employee Performance and Productivity dataset provides insights into workforce behavior and outcomes. Through AI-driven analysis, we aim to identify performance trends, evaluate productivity drivers, and support evidence-based talent management strategies.

By combining these three datasets, this project highlights the cross-functional value of AI and its potential to streamline workflows, personalize services, and reduce operational costs. Each dataset represents a unique business scenario where AI can assist in making faster, data-informed decisions. The project ultimately seeks to demonstrate that AI is not just a technological trend but a fundamental enabler of modern business intelligence.

II. Dataset Overview

No	Dataset Name	Domain	No. of Records	No. of Features	Model Applied	Missing Value?
1	Bank Marketing Campaign Data	Marketing	45,211	16	Classification	No
2	Bike Sales Data	Retail	100,000	11	Regression	No

3	Employee Performance & Productivity	Human Resources	100,000	20	Classification	No
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III. Tools Used and Why

No	Tools	Reasons
1	Python (Google Colab)	Perfect for exploratory data analysis and cleaning, thanks to its free, cloud-based Jupyter environment with Python and libraries like Pandas, Matplotlib, and Seaborn. With GPU/TPU support, it handles large datasets (e.g., 100,000 bike sales records) efficiently. It enables custom scripts for handling missing data and uncovering patterns like sales trends by model or location.
2	Julius AI	Julius AI’s natural language interface makes ARIMA forecasting simple and accessible. It’s ideal for quick, visual revenue predictions without deep coding. For stationary data, Julius fits ARIMA models and auto-generates intuitive plots—great for rapid prototyping or non-technical users.
3	Deepnote	Deepnote offers a collaborative space for thorough ARIMA model evaluation. It supports statistical checks (e.g., residuals, AIC/BIC), rich visualizations, and team workflows. Its power with large datasets makes it ideal for uncovering sales drivers and comparing model performance in a professional format.

IV. Key Findings (with Visuals)

1. Descriptive statistics by Python

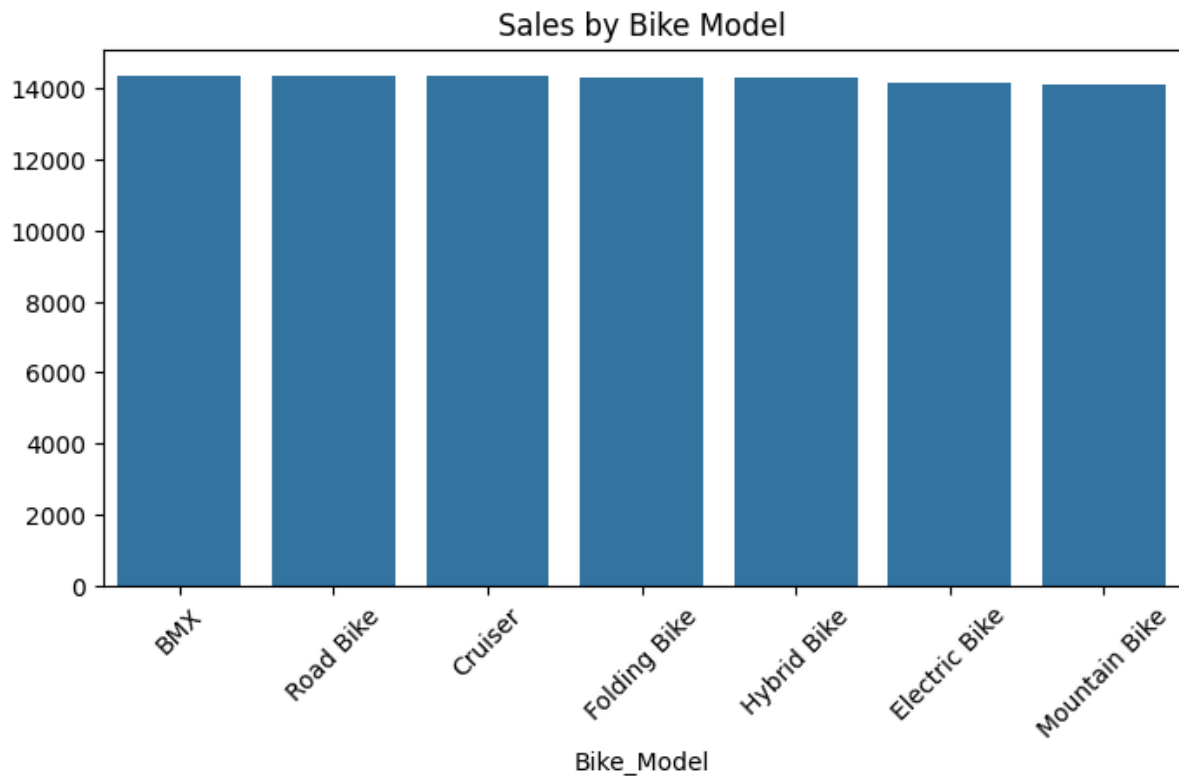
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[ ] df.describe()
```

	Sale_ID	Customer_ID	Price	Quantity	Salesperson_ID	Customer_Age
count	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000
mean	50000.500000	5495.450720	2598.182340	2.997110	549.906300	44.043410
std	28867.657797	2598.873948	1384.943133	1.414235	259.679221	15.313761
min	1.000000	1000.000000	200.010000	1.000000	100.000000	18.000000
25%	25000.750000	3249.000000	1399.787500	2.000000	324.000000	31.000000
50%	50000.500000	5491.000000	2598.575000	3.000000	550.000000	44.000000
75%	75000.250000	7738.000000	3795.982500	4.000000	775.000000	57.000000
max	100000.000000	9999.000000	4999.810000	5.000000	999.000000	70.000000

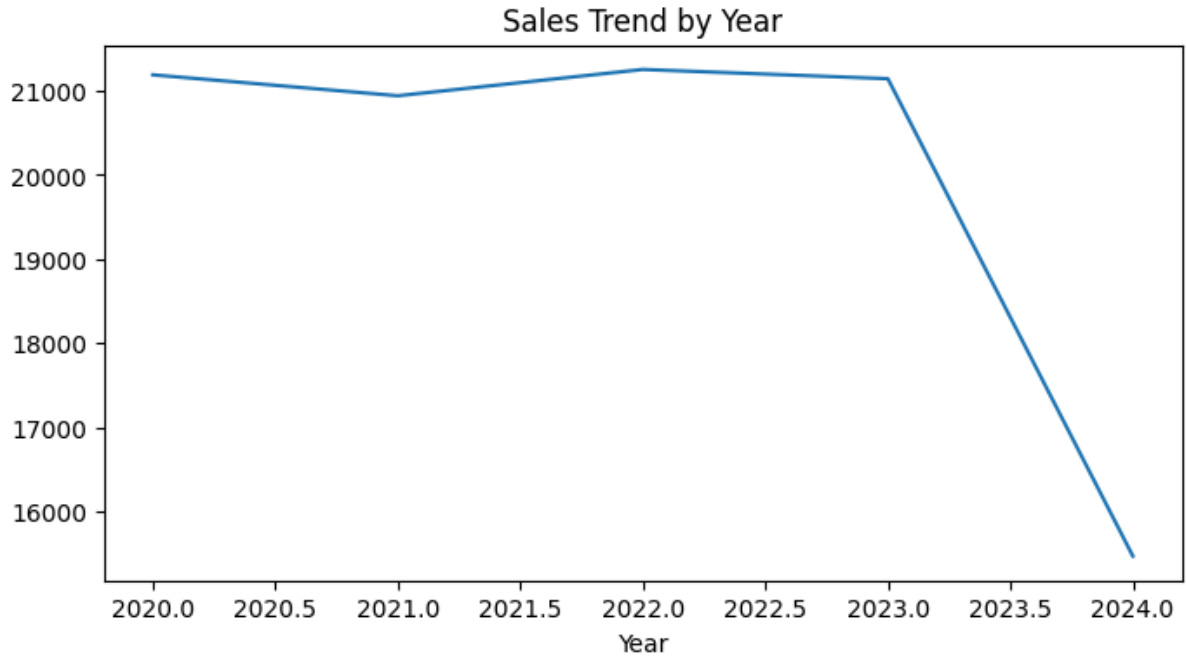
Insight: The average price stands at \$2,598 with a standard deviation of \$1,385, ranging from \$200 to \$4,999, and shows a right-skewed distribution where 25% of sales are below \$1,400, 50% below \$2,599, and 75% below \$3,796, with the majority clustered between

\$1,400 and \$3,796. Quantity sold averages 3 units per sale, with a standard deviation of 1.41 and a range of 1 to 5, reflecting an even distribution where 25% involve 2 units, 50% involve 3, and 75% involve 4 or more, highlighting a tendency for bulk purchases. Customer_Age averages 44 years with a standard deviation of 15.3, ranging from 18 to 70, and shows a balanced demographic with 50% of customers aged 31 to 57.

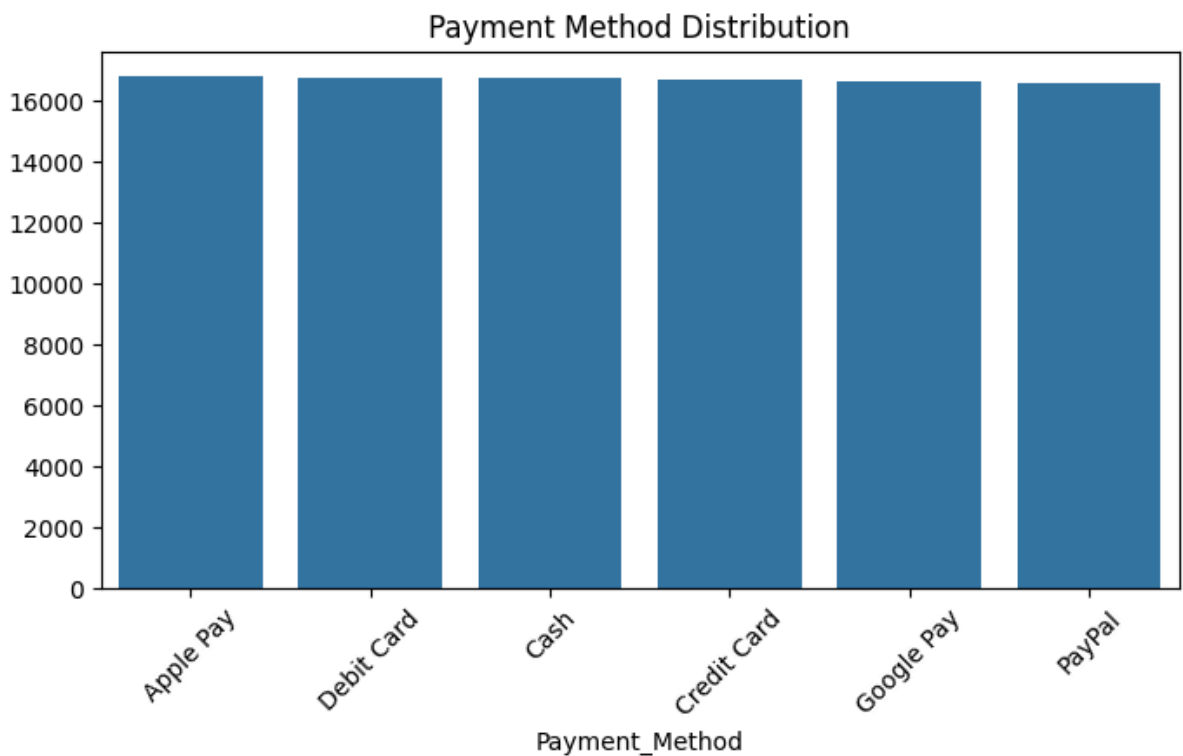
2. *Exploratory data analysis by Python*



Insight: The bar chart shows sales distribution by Bike_Model. All models (BMX, Road Bike, Cruiser, Folding Bike, Hybrid Bike, Electric Bike, Mountain Bike) have similar sales, ranging from approximately 12,000 to 14,000 units. No single model dominates, suggesting balanced demand across categories. Focus inventory and marketing evenly across all models.



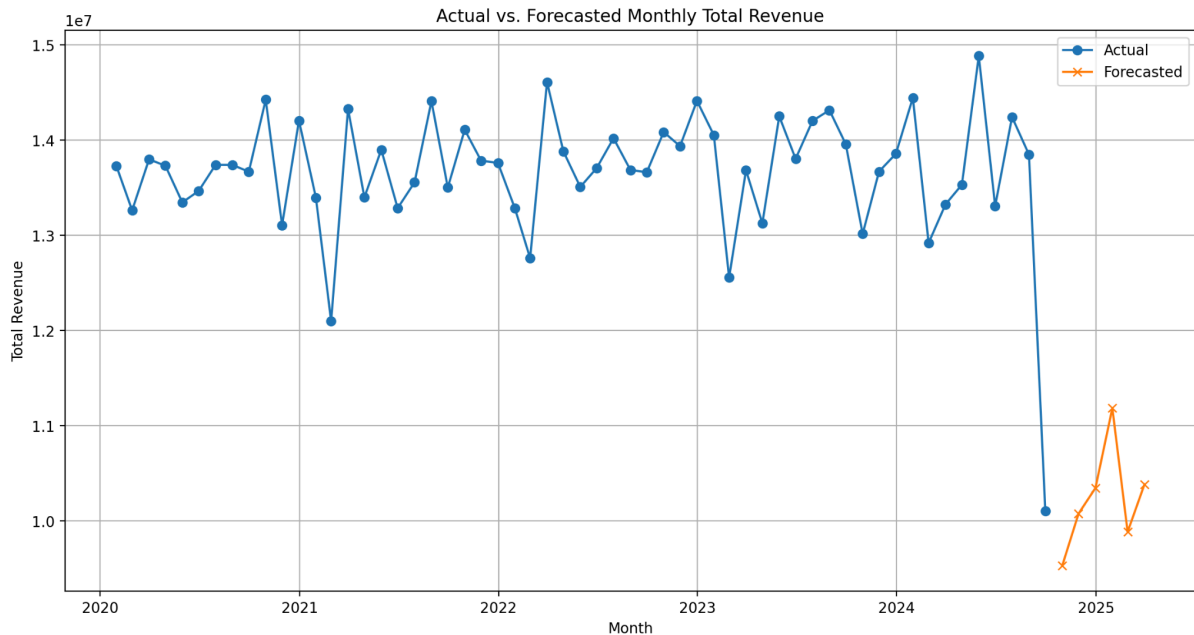
Insight: The line chart shows sales trends by year from 2020 to 2024. Sales peaked at ~21,000 in early 2020, stabilized around 20,000 from 2021 to mid-2022, then declined sharply to ~16,000 by mid-2024. Indicates a downward trend since 2023, suggesting market saturation or external factors.



Insight: The bar chart shows payment method distribution. All methods (Apple Pay, Debit Card, Cash, Credit Card, Google Pay, PayPal) have similar usage, ranging from ~14,000 to

~16,000 transactions. Indicates no strong preference, suggesting robust payment infrastructure.

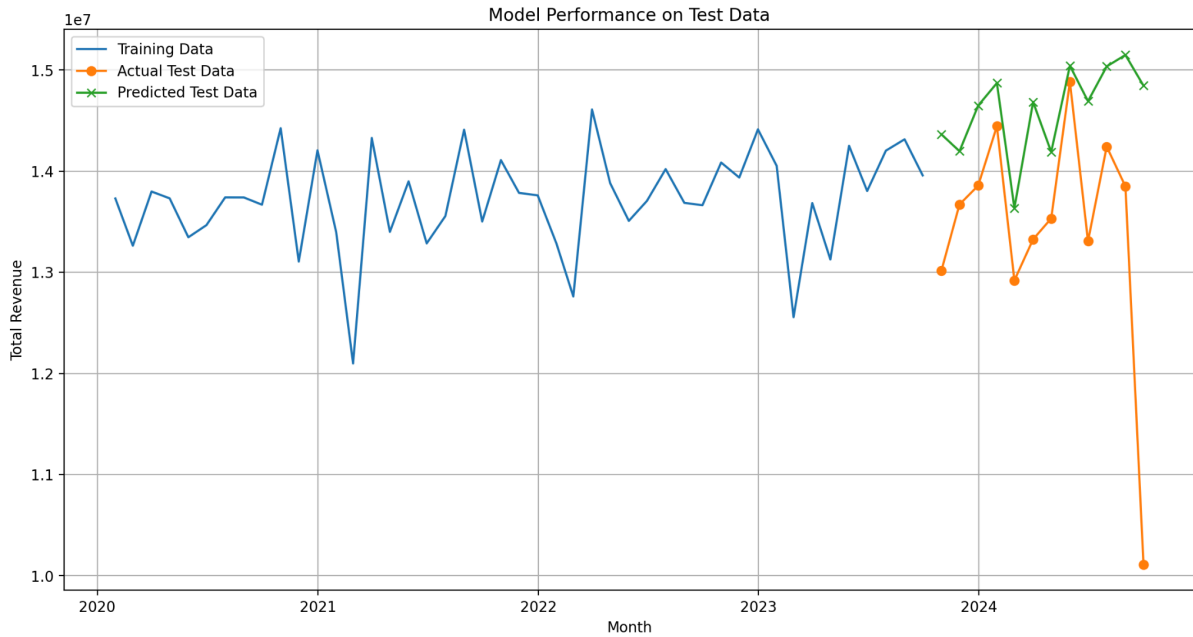
3. ARIMA forecasting model by Julius AI



Insight: The line chart compares actual vs. forecasted monthly total revenue (in millions). Actual revenue (blue) fluctuated between 1.3M and 1.5M from 2020 to mid-2024, then dropped sharply to 1.0M. Forecasted revenue (orange) starts in 2024, predicting a rise from 1.0M to 1.1M by mid-2025.

The forecasted monthly total revenue for the next 6 months has been generated using the SARIMAX model, which accounts for both autoregressive and seasonal components. The plot shows the historical monthly revenue along with the forecasted values, giving a visual sense of expected future trends.

- The forecasted values are as follows:
- October 2024: approximately 9.53 million
- November 2024: approximately 10.08 million
- December 2024: approximately 10.35 million
- January 2025: approximately 11.19 million
- February 2025: approximately 9.89 million
- March 2025: approximately 10.39 million



Insight: The line chart shows ARIMA model performance on test data. Training data (blue) fluctuates between 1.3M and 1.5M from 2020 to mid-2024. Actual test data (orange) and predicted test data (green) align closely from mid-2024, with actual dropping to 1.0M and predicted showing a slight rise. A good fit on a test set indicates reliable forecasting, but the recent decline suggests external factors.

4. Model evaluation by Deepnote

4.1 Apply Augmented Dickey-Fuller test to check stationary

Augmented Dickey-Fuller (ADF) Statistics				
ADF Statistic	p-value	Lags Used	Number of Observations	Critical Value (1%)
-4.761264261	6.45E-05	3	53	-3.560242359
				Critical Value (5%)
				-2.917850207
				Critical Value (10%)
				-2.596796415

- No Differencing Required

The data is already stationary, so the integration order for ARIMA would be $d = 0$.

- Modeling Implications

Since the series is stationary, AR, MA, or ARMA components can be applied directly. If seasonality is detected, SARIMA with seasonal terms can be used.

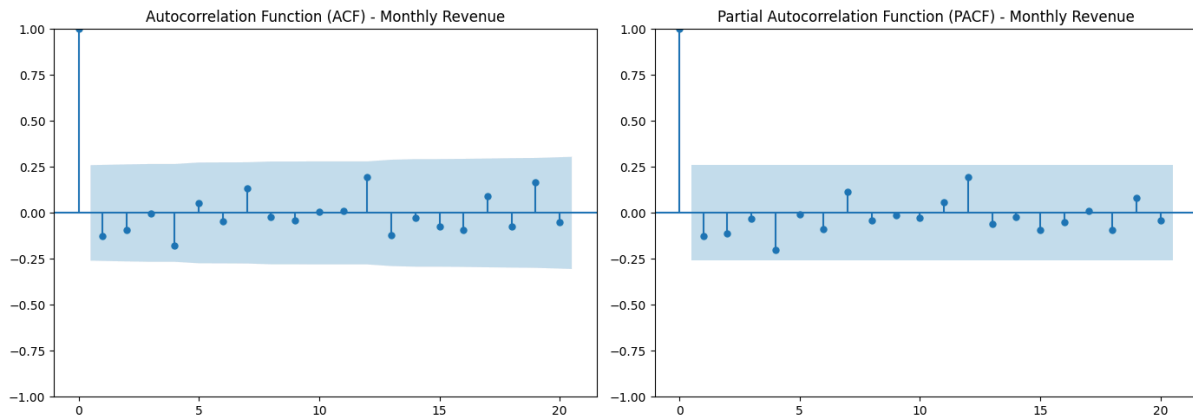
- Forecast Stability

Stationarity implies that statistical properties like mean and variance remain stable over time, increasing the reliability of model forecasts.

- Potential for Direct Analysis

With no transformation needed to achieve stationarity, analysis and forecasting can proceed immediately, saving preprocessing time.

4.2 ACF and PACF plots



ACF Insights (Left Plot)

- Significant spikes at **lag 1** and smaller but notable at **lag 2–3**, then it tapers off.
- This suggests a possible **MA(1)** or **MA(2)** component.

PACF Insights (Right Plot)

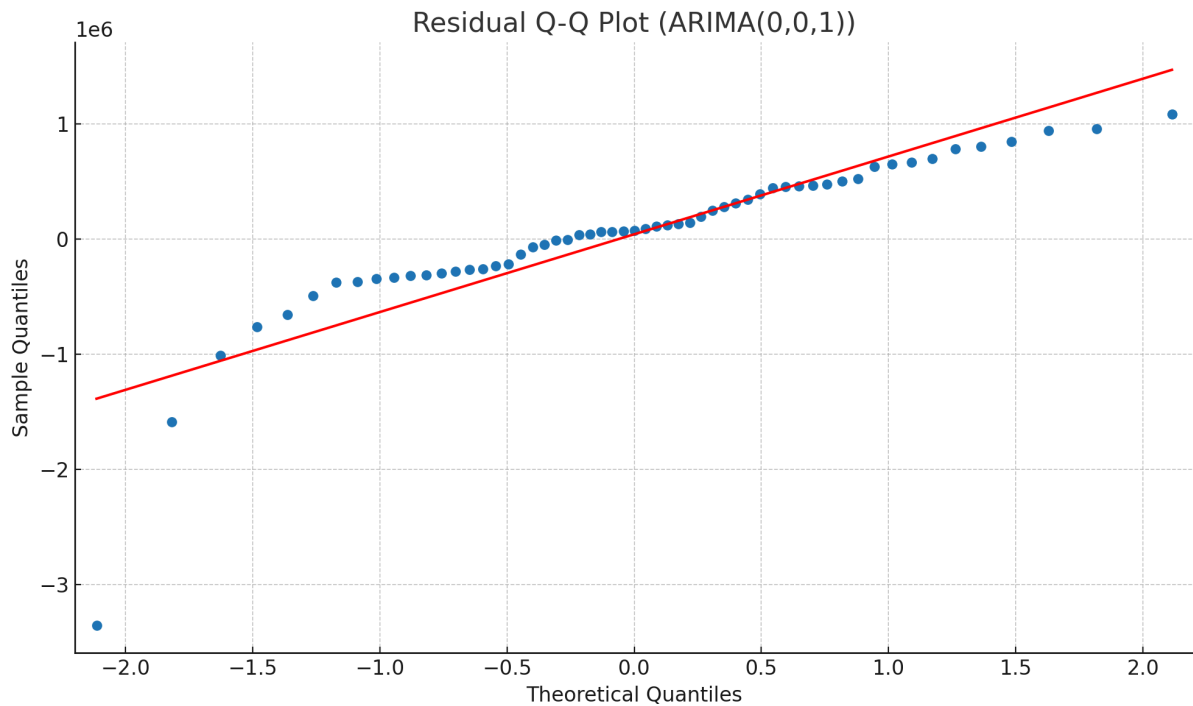
- Strong spike at **lag 1**, and smaller ones at **lag 2–3** before dropping toward zero.
- This pattern often indicates an **AR(1)** or possibly **AR(2)** structure.

4.3 Model fitting and selection by AIC/BIC comparison

Order (p,d,q)	AIC	BIC
(0,0,1)	1698.65	1704.77
(1,0,1)	1698.93	1707.10
(1,0,0)	1699.86	1705.99

Key Insights:

- **ARIMA(0,0,1)** (pure MA(1) model) has the **lowest AIC and BIC**, making it the best fit statistically.
- **ARIMA(1,0,1)** is a close second, suggesting there may be a minor AR(1) effect, but it's slightly more complex without clear improvement in fit.
- All top models confirm **no differencing needed (d=0)** — consistent with your ADF test result.



Insight: Some tail heaviness (a few large shocks), but broadly close to normal.

V. Tool Comparison

In terms of usability, Google Colab offers moderate user-friendliness but is primarily suited for individuals with Python programming knowledge. Its extensive capabilities make it powerful for experienced coders, but less approachable for beginners. Julius AI, by contrast, is designed to be highly intuitive, catering to non-technical users by enabling natural language-based commands and simple workflows. Deepnote presents a balance between the two, offering an accessible interface while also supporting collaborative features for teams with mixed skill levels.

Regarding computational speed, Google Colab stands out by providing GPU and TPU acceleration, enabling high-performance data processing for resource-intensive tasks. Julius AI is optimized for quick, straightforward analyses, offering near-instant results for smaller-scale workloads. Deepnote maintains reasonable performance for most tasks but can be slower when handling large-scale computations compared to Google Colab's hardware acceleration capabilities.

Visualization capacity also varies significantly across the platforms. Google Colab supports robust, customizable visualizations by integrating with popular Python libraries such as Matplotlib, Seaborn, and Plotly, giving users full creative control. Julius AI offers simple, automatically generated visual outputs, which are suitable for basic reporting but lack advanced customization. Deepnote, meanwhile, delivers advanced and interactive visualization capabilities, making it particularly useful for collaborative analytical projects where presentation quality is important.

Flexibility is another key differentiator. Google Colab provides the highest level of flexibility, allowing full custom coding and integration with a wide range of Python packages. Julius AI's flexibility is limited by its reliance on predefined functions, which restricts the scope of more specialized analyses. Deepnote offers a middle ground, supporting both coding and guided analytics while enhancing teamwork through real-time collaboration features.

User experience and ease of learning further influence tool selection. Google Colab's interface is functional but minimalistic, designed primarily for coders who value control over aesthetics. Julius AI offers a seamless, beginner-friendly experience by focusing on accessibility through natural language interaction. Deepnote provides a polished and professional user interface, well-suited for team-based work. In terms of learning curve, Google Colab can be challenging for non-programmers due to its dependence on Python, whereas Julius AI requires minimal training and can be used effectively with little guidance. Deepnote sits between these extremes, with a moderate learning curve eased by tutorials and an intuitive design.

When considering suitability for task types, Google Colab is best for coding-based projects, such as Python development, data analysis, and machine learning workflows that require significant computational resources. Deepnote also supports coding but adds collaborative and project management capabilities, making it ideal for team-based coding environments. For no-code workflows, Julius AI emerges as the most suitable option, allowing users to conduct analysis and visualization entirely through natural language commands without the need for programming skills.

In conclusion, the choice between these platforms should be driven by the specific needs, technical skills, and project requirements of the user. Google Colab is the preferred tool for experienced programmers seeking computational power and flexibility. Julius AI is optimal for non-technical users who prioritize speed and ease of use in no-code environments. Deepnote offers an effective balance, combining coding capabilities with strong collaborative features, making it a valuable solution for team-oriented analytics projects. This comparative evaluation provides a framework for aligning tool selection with analytical goals in both academic and professional contexts.

VI. Business Insights & Conclusion

1. Practical applications: improving marketing campaigns, optimizing inventory & staffing decisions.

The forecasting and analytical approaches used in this study present several practical applications for business decision-making, particularly in the areas of marketing, inventory management, and staffing optimization.

First, in improving marketing campaigns, ARIMA-based forecasts can be utilized to identify high-demand bike models and adjust promotional strategies accordingly. For example, evenly distributed sales across models may suggest the need for balanced promotion, while a decline in sales observed in 2024 could be addressed through targeted incentives or discounts.

Additionally, the distribution of payment methods provides opportunities for multi-channel marketing, allowing businesses to design campaigns that align with customer purchasing preferences.

Second, in optimizing inventory decisions, stock levels can be managed based on ARIMA predictions and observed sales trends. This approach allows businesses to prepare for projected revenue declines, such as the anticipated dip between 2024 and 2025, by adopting a just-in-time inventory system. Such a strategy minimizes overstocking risks, particularly when demand is balanced across multiple models, and helps maintain operational efficiency.

Third, staffing decisions can be aligned with forecasted sales fluctuations. For instance, the early 2022 sales peak may require increased staff availability, while anticipated declines in 2024–2025 could justify reduced staffing levels. Furthermore, scheduling adjustments can be made according to store location performance, ensuring that labor resources are allocated proportionally to sales volume, thereby optimizing labor costs without compromising service quality.

2. Limitations and future work.

Despite the value of these findings, several limitations must be acknowledged. ARIMA models rely on the assumption of stationarity, which may not hold in the presence of sustained sales declines. The dataset also lacks external explanatory factors, such as macroeconomic conditions or industry-specific events, that could provide more context to observed trends. Furthermore, the initial forecasting process did not include a dedicated test set for validation, which reduces the reliability of the model's predictive accuracy. Finally, seasonality patterns were not explored in depth, which may have overlooked periodic fluctuations in sales.

Future research should address these limitations by incorporating additional variables, such as holiday effects, weather conditions, and macroeconomic indicators, through models like SARIMA or Facebook Prophet. Expanding the dataset to include larger test sets would improve forecast validation. Additionally, advanced machine learning techniques, such as Long Short-Term Memory (LSTM) networks, could be explored to capture non-linear patterns in sales data. Enhancing the dataset with customer feedback and demographic information would also enable deeper insights into demand drivers and consumer behavior.

3. How these tools help in businesses

The three platforms evaluated—Google Colab, Julius AI, and Deepnote—offer distinct contributions to business analytics. Google Colab enables data-driven decision-making through customizable coding environments, allowing for in-depth trend analysis and price optimization. Julius AI simplifies forecasting for non-technical personnel, enabling rapid revenue predictions that can directly inform marketing and inventory adjustments. Deepnote enhances collaborative analysis, enabling teams to jointly evaluate models, refine operational

strategies, and derive actionable insights such as staffing adjustments across multiple locations.

By leveraging these tools appropriately, businesses can transform raw data into strategic actions, supporting competitive advantage in dynamic market environments.